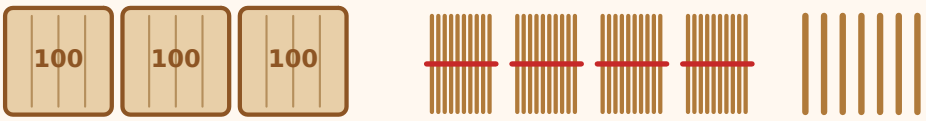
Chapter 1

Arithmetic of whole numbers

the four things you can do to numbers, done with sticks, lifts and dots



3 hundreds 4 tens 7 ones

347

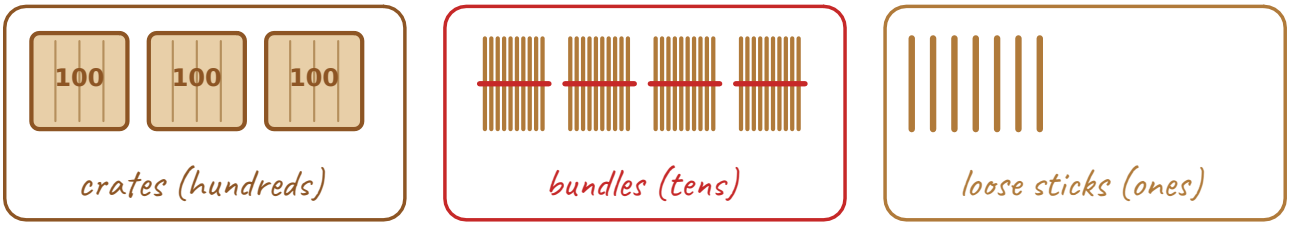
Everything in maths is built on four actions: adding, taking away, multiplying and dividing. You already do these every day with money. This chapter slows them right down so you can **see** what is happening inside them, because every later chapter, from fractions to calculus, is these four actions in disguise.

It also covers negative numbers, the order you do things in (the famous BODMAS), and the factor and multiple ideas that fractions (Chapter 2) depend on.

Goes with **Lecture 1** (mathematical writing and BODMAS).

Numbers are bundles of ten

Why does 347 mean three hundred and forty-seven? Imagine counting sticks. Every time you reach **ten sticks**, you tie them into a **bundle**. Every time you have **ten bundles**, you pack them in a **crate** (a crate holds 100 sticks).



A written number is just a report of what is in each box, in order: 3 crates, 4 bundles, 7 loose sticks is written 347. The **position** of a digit tells you its size. That is called **place value**.

Zero is a place-holder. 307 means 3 crates, **no** bundles, 7 sticks. Without the 0 you'd write 37, which is a completely different number. Zero says "this box is empty".

A car mileage counter works the same way. When the ones wheel goes past 9, it rolls back to 0 and pushes the tens wheel on by one:

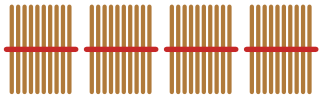


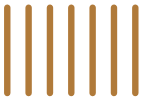
19 plus 1: the ones wheel rolls from 9 to 0 and the tens wheel clicks from 1 to 2. That "roll over and push the next one" is exactly what **carrying** means in adding.

Adding: put the piles together, re-bundle

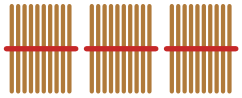
Work out $47 + 38$. Put 47 sticks and 38 sticks side by side, each in its columns:

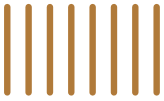
47





38





ones: 7 + 8 = 15 loose sticks. That's more than ten, so tie ten of them into a new bundle: 15 = 1 bundle and 5 sticks

Step by step, column by column, starting with the ones:

$$\begin{array}{r} {\scriptstyle 1} \\ 47 \\ +\,38 \\ \hline 85 \end{array}$$

1. ones column: 7 + 8 = 15. Write the 5, and carry the new bundle (1 ten) over to the tens column.

2. tens column: 4 + 3 = 7 bundles, plus the 1 carried bundle = 8 bundles. Write 8.

3. So 47 + 38 = 85 (8 bundles and 5 sticks).

Carrying is just re-bundling: ten ones become one ten, ten tens become one hundred. Adding also doesn't care about order: $47 + 38 = 38 + 47$.

Taking away: sometimes you must open a bundle

Work out $52 - 17$. Start with 52: five bundles and 2 loose sticks. You need to take away 7 loose sticks... but you only have 2!



52 = 5 bundles, 2 sticks

open one bundle: now 4 bundles and 2 + 10 = 12 loose sticks

$$\begin{array}{r} {\scriptstyle 4\ \ 12} \\ 5\!\!\not{2} \\ -\ 17 \\ \hline 35 \end{array}$$

1. ones: 2 - 7? Not enough. Open a bundle: the 5 tens become 4, the 2 ones become 12.

2. ones: 12 - 7 = 5. tens: 4 - 1 = 3.

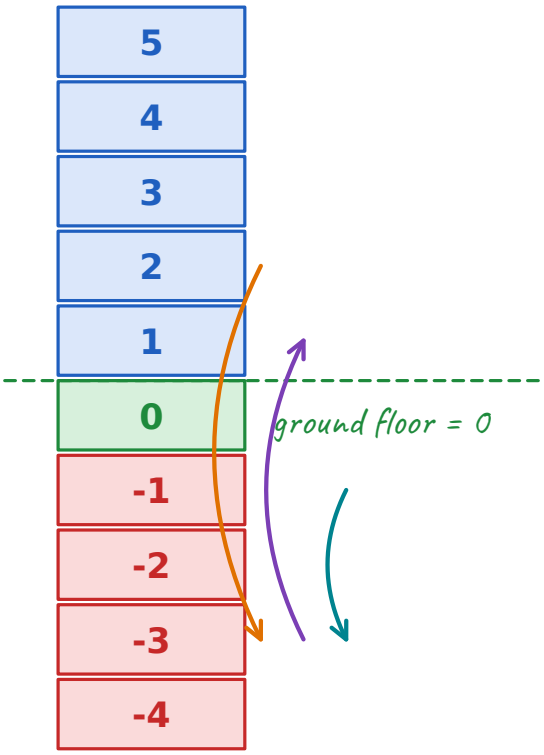
3. So 52 - 17 = 35.

Check by adding back: $35 + 17 = 52$. ✓ Taking away is the undo button for adding.

Order matters here! $52 - 17$ is not the same as $17 - 52$. (The second one goes below zero, which is the next idea.)

Below zero: the lift in a building

A lift has floors above the ground (1, 2, 3...) and basement floors below it. Call the ground floor 0 and the basement floors **negative**: -1, -2, -3 (said "minus one" or "negative one").



$$2 - 5 = -3$$

start at 2, go DOWN 5 floors

$$-3 + 4 = 1$$

start at -3, go UP 4 floors

$$-1 - 2 = -3$$

start at -1, go DOWN 2 floors

Adding a positive number moves you **up**; **taking away** a positive number moves you **down**. The rules below cover every other case. The **size** of a number is how far it is from 0, ignoring the sign: the size of -7 is 7.

RULE: adding and subtracting with negative numbers

Adding a negative is the same as subtracting: $a + (-b) = a - b$.
E.g. $7 + (-3) = 7 - 3 = 4$.

Subtracting a negative is the same as adding: $a - (-b) = a + b$.
E.g. $5 - (-2) = 5 + 2 = 7$.

Two signs, different: subtract the smaller size from the bigger size, and give the answer the **sign of the number with the bigger size**. E.g. $-9 + 4 = -5$ (9 is bigger, it was negative); $-3 + 8 = 5$ (8 is bigger, it was positive).

Both negative: add the sizes, the answer is negative. E.g. $-4 - 6 = -10$, $-4 + (-6) = -10$.

$$7 + (-3) = 7 - 3 = 4$$ ← + (−) becomes −

$$-9 + 4 = -5$$ ← sizes 9 and 4: 9 − 4 = 5; the 9 was negative

$$-2 - (-5) = -2 + 5 = 3$$ ← − (−) becomes +, then sizes 5 − 2 = 3, sign of the 5

$$-4 + (-6) = -4 - 6 = -10$$ ← both negative: add sizes, keep −

Why subtracting a negative adds. Think of money: -2 is a debt of £2. If someone **takes away** a debt you owe, you are **better off** by £2, so $5 - (-2) = 7$. Two minus signs **next to each other** make a plus.

Multiplying and dividing negatives

RULE: signs when multiplying and dividing

Work out the sizes as usual, then fix the sign.

Same signs give a positive answer: $(+) \times (+) = +$ and $(-) \times (-) = +$.

Different signs give a negative answer: $(+) \times (-) = -$ and $(-) \times (+) = -$.

Division follows exactly the same sign rules as multiplication.

Several numbers: count the minus signs. **Even** count gives $+$, **odd** count gives $-$. E.g. $(-2) \times (-3) \times (-1) = -6$ (three minus signs).

Signs	×	÷	Answer
$+$ and $+$	$3 \times 4 = 12$	$12 \div 3 = 4$	**positive**
$+$ and $-$	$3 \times (-4) = -12$	$12 \div (-3) = -4$	**negative**
$-$ and $+$	$(-3) \times 4 = -12$	$-12 \div 3 = -4$	**negative**
$-$ and $-$	$(-3) \times (-4) = 12$	$-12 \div (-3) = 4$	**positive**

Why is negative times negative positive? Watch a pattern. Each line below multiplies by one less than the line above, so the answer changes by 3 each time:

$3 \times 2 = 6$

$3 \times 1 = 3$

$3 \times 0 = 0$

$3 \times (-1) = -3$

$3 \times (-2) = -6$

down 3 each time

$-3 \times 2 = -6$

$-3 \times 1 = -3$

$-3 \times 0 = 0$

$-3 \times (-1) = 3$

$-3 \times (-2) = 6$

UP 3 each time

On the left the answers go 6, 3, 0, then keep going down: -3, -6. On the right they go -6, -3, 0, then keep going **up**: 3, 6. The pattern forces it: a negative times a negative is **positive**.

Careful: the "two minuses make a plus" rule for × and ÷ is about **two numbers**. For adding it is different: $-3 + (-4) = -7$, not $+7$.

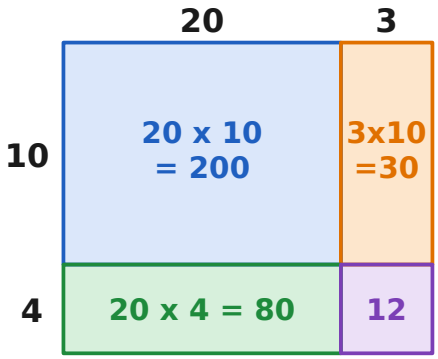
Multiplying: rows of dots

4×6 means "4 lots of 6". Draw it as 4 rows of 6 dots: a rectangle.



4 rows of 6 = 24 dots.
Turn the page sideways and it's
6 rows of 4: still 24. Order doesn't matter.

Bigger numbers: split the rectangle. 23×14 is a rectangle 23 wide and 14 tall. Split 23 into 20 + 3 and 14 into 10 + 4. Now it's four easy rectangles (the **grid method**):



Add the four pieces:
200 + 30 + 80 + 12 = 322

$$23 \times 14 = 322$$

RULE: laws of multiplication (and which operations can be swapped)

Commutative: order does not matter for $\times$ and $+$: $a \times b = b \times a$, $a + b = b + a$. **It does matter for $-$ and $\div$:** $7 - 3 \neq 3 - 7$, $20 \div 4 \neq 4 \div 20$.

Distributive law: $a(b + c) = ab + ac$. Multiplying a sum means multiplying **each part** and adding. The grid method above is this law: $23 \times 14 = 23 \times 10 + 23 \times 4 = 230 + 92 = 322$.

Also works with minus: $a(b - c) = ab - ac$. E.g. $6 \times 19 = 6 \times 20 - 6 \times 1 = 114$.

This "split into easy rectangles" picture comes back in Chapter 11 , where it's used to multiply out brackets like $(x + 3)(x + 2)$.

Times tables. Quick recall of the times tables up to 10 x 10 makes everything later much less tiring. If they are shaky, the grid on the next page is worth pinning up.

The times-table square

Find a row and a column; where they meet is the answer. The diagonal holds the **square numbers** (1×1, 2×2, ...), which come up constantly later.

x	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

square numbers on the diagonal

Dividing: sharing out, and what's left over

$20 \div 4$ asks: **share 20 sweets between 4 people. How many each?** Deal them out one at a time, like cards:



5 each

Division is the undo button for multiplying: $4 \times 5 = 20$, so $20 \div 4 = 5$. The same sum can be asked the other way round: "how many groups of 4 can I make from 20?" (also 5).

RULE: division

Division undoes multiplication: if $a \times b = c$ then $c \div b = a$ and $c \div a = b$. Use this to check every division.

Order matters: $20 \div 4 = 5$ but $4 \div 20 = \frac{4}{20} = \frac{1}{5}$. Division is not commutative.

With a remainder: number = quotient × divisor + remainder, and the remainder is always smaller than the divisor. E.g. $23 = 5 \times 4 + 3$.

You can never divide by 0.

Remainders. Share 23 sweets between 4: each gets 5 (that uses 20), and **3 are left over**. We write $23 \div 4 = 5$ remainder 3, or "5 r 3".

Bigger division, sharing crate by crate ("short division"): $156 \div 6$.

$$\begin{array}{r} 26 \\ 6\overline{)\,15^{3}6} \end{array}$$

1. 6 into 1 (hundred)? Can't. Look at 15 (tens) instead.
2. 6 into 15 = 2, remainder 3. Write 2 on top.
   The 3 tens left over become 30 ones: write a small 3.
3. 6 into 36 = 6. Write 6. So 156 ÷ 6 = 26.

Check by multiplying back: $26 \times 6 = 120 + 36 = 156$. ✓

Dividing by zero is not allowed. "Share 5 sweets between 0 people" has no sensible answer. Calculators give an error, and so does maths.

The order of operations (BODMAS)

PROCEDURE: order of operations

1. **Brackets** first. With brackets inside brackets, work from the **innermost** outwards.
2. **Orders**: powers and roots (5^2, $\sqrt{9}$).
3. **Division and Multiplication**: **equal priority**. Work **left to right**, doing whichever comes first.
4. **Addition and Subtraction**: **equal priority**. Work **left to right**.

RULE: hidden brackets

A **fraction bar** acts as brackets round the whole top and the whole bottom: $\frac{4+8}{5-2} = (4+8) \div (5-2)$.

A power applies only to what it touches: $-3^2 = -(3^2) = -9$, but $(-3)^2 = (-3) \times (-3) = 9$.

Like getting dressed (socks before shoes), doing the steps in a different order gives a different result. The order is often remembered as **BODMAS**:

B **Brackets**
( ) first, always

O **Orders**
powers and roots, like 5^2 or $\sqrt{9}$

DM **Division and Multiplication**
equal rank: work left to right

AS **Addition and Subtraction**
equal rank: work left to right

(Some people learned **BIDMAS** or **PEMDAS**: same rules, different words.) Note that D and M are **equal**: do whichever comes first reading left to right. Same for A and S.

Example, one step per line: $3 + 4 \times 2^2 - (6 - 1)$

$$3 + 4 \times 2^2 - (6 - 1) \quad \longleftarrow \textit{B: brackets first: 6 - 1 = 5}$$

$$3 + 4 \times 2^2 - 5$$ ← O: powers: 2 squared = 4

$$3 + 4 \times 4 - 5$$ ← M: multiply: 4 x 4 = 16

$$3 + 16 - 5$$ ← A/S left to right: 3 + 16 = 19

$$19 - 5 = 14$$ ← then 19 - 5

Equal priority means left to right:

$$12 \div 3 \times 2 = 12 \div 3 \times 2 = 4 \times 2 = 8$$

÷ comes first reading left to right

$$10 - 4 + 3 = 10 - 4 + 3 = 6 + 3 = 9$$

– comes first reading left to right

Nested brackets, innermost first: $2 \times [3 + (8 - 2)^2]$

$$2 \times [3 + (8 - 2)^2] = 2 \times [3 + 6^2]$$ ← inner bracket: 8 – 2 = 6

$$= 2 \times [3 + 36] = 2 \times 39 = 78$$ ← power, then the outer bracket

Fraction bar and powers of negatives:

$$\frac{4 + 8}{5 - 2} = \frac{12}{3} = 4$$ ← top and bottom first, then divide

$$(-3)^2 = 9, \qquad -3^2 = -9$$ ← the square touches –3 in the first, only 3 in the second

✗ Common mistakes

✗ $12 \div 3 \times 2 = 12 \div 6 = 2$ ✓ $12 \div 3 \times 2 = 4 \times 2 = 8$

× is not done before ÷: they are equal, so go left to right

✗ $10 - 4 + 3 = 10 - 7 = 3$ ✓ $10 - 4 + 3 = 6 + 3 = 9$

same for + and –: left to right

✗ $-3^2 = 9$ ✓ $-3^2 = -(3^2) = -9$

without brackets the power does not include the minus sign

The internet argument: 8 ÷ 2(2+2)

Your Lecture 1 slides show this famous puzzle. People get **16** or **1**, and argue for days. Here are the two readings side by side:

Reading A: strict left to right

$$8 \div 2(2+2)$$

$$8 \div 2 \times 4$$

$$4 \times 4$$

$$16$$

brackets first, then ÷ and x left to right

Reading B: "2(2+2)" is one lump

$$8 \div 2(2+2)$$

$$8 \div (2 \times 4)$$

$$8 \div 8$$

$$1$$

treat 2(4) as a single number, because it's glued

The honest answer: the expression is **badly written**. The rules above give 16, but plenty of calculators and scientists treat "$2(2+2)$" as glued together and give 1. The lesson from your lecture is: **when you write maths, remove the doubt.** Use extra brackets, and use a fraction line instead of ÷:

$$\frac{8}{2}(2+2) = 16 \qquad \frac{8}{2(2+2)} = 1$$

Nobody can argue with either of those. That's why, from now on, this book (and your lecturer) almost never uses the ÷ sign once letters appear.

Factors: every way to make a rectangle

A **factor** of 12 is a number that divides into 12 exactly, with no remainder. Here's a picture of it: take 12 dots and find **every rectangle** you can make with them.



1 x 12



2 x 6



3 x 4

(4 x 3, 6 x 2 and 12 x 1 are the same rectangles turned sideways)

The side lengths are the factors: **1, 2, 3, 4, 6, 12**. Factors come in pairs that multiply to 12: 1 & 12, 2 & 6, 3 & 4.

Primes are the numbers bigger than 1 that **only** make a single line of dots:

7 dots: only 1 x 7 works. No other rectangle. 7 is prime.



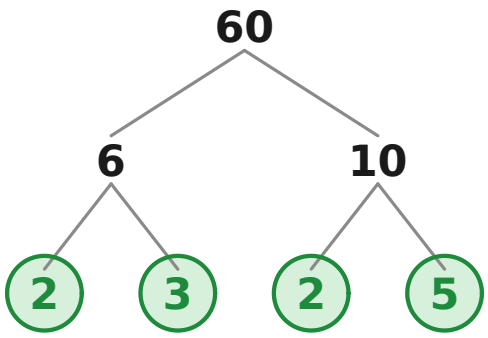
8 dots makes 2 x 4: 8 is NOT prime

The first primes are **2, 3, 5, 7, 11, 13, 17, 19, 23, 29**. (1 is not counted as prime.) A whole number bigger than 1 that isn't prime is called **composite** (it has a factor other than 1 and itself). 0 and 1 are neither prime nor composite.

Multiples are the times table of a number: the multiples of 4 are 4, 8, 12, 16, 20... "12 is a multiple of 4" and "4 is a factor of 12" say the same thing from opposite ends.

Breaking a number into primes: a factor tree

Every whole number bigger than 1 can be built by multiplying primes together, and in only one way. To find the primes, keep splitting into any factor pair until only primes are left. Take 60:



60 = 6 x 10

6 = 2 x 3, 10 = 2 x 5

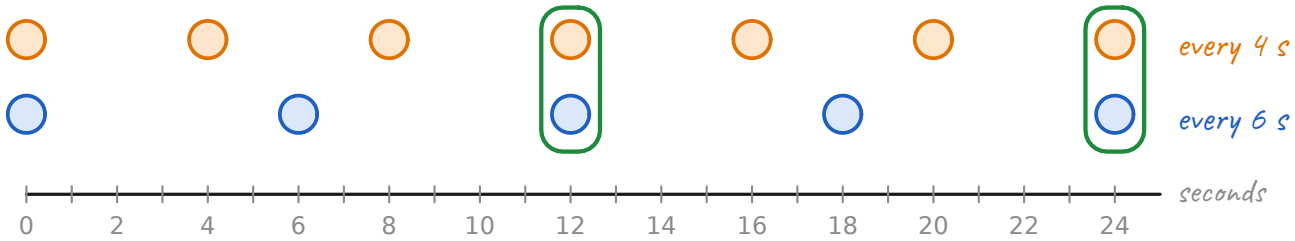
circled numbers are prime: stop

$$60 = 2 \times 2 \times 3 \times 5$$
$$= 2^2 \times 3 \times 5$$

You could start with $60 = 4 \times 15$ instead; you'd still end with the same primes. This list of primes is the number's **prime factorisation**.

LCM: when do two blinking lights blink together?

One traffic light blinks every 4 seconds; another blinks every 6 seconds. They both blink at the start. When do they next blink **at the same time**?



Together at **12 seconds** (and again at 24, 36...). 12 is the smallest number in **both** times tables: the **Lowest Common Multiple**, or **LCM**, of 4 and 6.

Chapter 2 uses the LCM to find the smallest piece-size two fractions can be cut into.

HCF: the biggest number that fits into both

The **Highest Common Factor** (HCF) of 12 and 18 is the **biggest** number that divides into both. List the factors and find the biggest one they share:

Factors of 12: 1, 2, 3, 4, **6**, 12

Factors of 18: 1, 2, 3, **6**, 9, 18

HCF = 6

Chapter 2 uses the HCF to simplify fractions in one step: $\frac{12}{18}$ becomes $\frac{2}{3}$ by dividing top and bottom by 6.

Don't mix them up: **M**ultiples get bigger (LC**M** is at least as big as both numbers). **F**actors get smaller (HC**F** is no bigger than either number).

HCF and LCM from prime factorisations

Listing factors or multiples is slow for big numbers. Prime factorisations give both in one go.

PROCEDURE: HCF and LCM using primes

1. Write each number as a product of primes, using powers: $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$.
2. **HCF**: take only the primes that appear in **every** number, each with its **lowest** power.
3. **LCM**: take **every** prime that appears in any of the numbers, each with its **highest** power.
4. Multiply out. Check: for two numbers, $\text{HCF} \times \text{LCM} =$ the two numbers multiplied together.

Example: HCF and LCM of 24 and 36.

$$24 = 2^3 \times 3, \qquad 36 = 2^2 \times 3^2$$ ← *factor trees*

$$\text{HCF} = 2^2 \times 3^1 = 12$$ ← *shared primes 2 and 3, lowest powers*

$$\text{LCM} = 2^3 \times 3^2 = 8 \times 9 = 72$$ ← *all primes, highest powers*

$$12 \times 72 = 864 = 24 \times 36 \checkmark$$ ← *the check*

Word problems: which one do I need? "Biggest thing that fits into both" (cutting, sharing into equal groups) is the **HCF**. "When do they next happen together" (timetables, repeating patterns) is the **LCM**.

Example. A floor is 84 cm by 60 cm. It is covered exactly with identical square tiles, as large as possible. What size are the tiles?

$$84 = 2^2 \times 3 \times 7, \qquad 60 = 2^2 \times 3 \times 5$$

tile side must divide both: HCF

$$\text{HCF} = 2^2 \times 3 = 12 \text{ cm}$$ ← *84 ÷ 12 = 7 tiles by 60 ÷ 12 = 5 tiles*

Example. Buses leave every 12 minutes and trams every 18 minutes. Both leave at 9:00. When do they next leave together?

$$12 = 2^2 \times 3, \quad 18 = 2 \times 3^2 \;\Rightarrow\; \text{LCM} = 2^2 \times 3^2 = 36$$

"together again": LCM

$$9{:}00 + 36 \text{ min} = 9{:}36$$

Find the missing number

PROCEDURE: finding an unknown

1. Write the unknown as $?$ (later in the book: a letter such as x).
2. **Undo** what was done to it, using the **inverse** operation: $+$ is undone by $-$, $-$ by $+$, $\times$ by $\div$, $\div$ by $\times$.
3. Apply the sign rules for negatives as you go.
4. **Check** by putting your answer back into the original line.

Example. $? - 7 = -3$

$$? = -3 + 7 = 4$$ ← undo − 7 by adding 7

$$4 - 7 = -3 \checkmark$$ ← check

Example. $-4 \times ? = 28$

$$? = 28 \div (-4) = -7$$ ← undo × (−4) by dividing; different signs: negative

$$-4 \times (-7) = 28 \checkmark$$ ← same signs: positive

Example. $? \div 3 = -6$

$$? = -6 \times 3 = -18$$ ← undo ÷ 3 by multiplying by 3

Example. The HCF of 12 and n is 4, and n is from 20 to 30. What could n be?

$$4 \mid n: \quad n \in \{20, 24, 28\}$$ ← 4 must divide n: multiples of 4 in range

$$\text{HCF}(12, 24) = 12 \neq 4$$ ← 24 shares the factor 3 with 12 as well: rule it out

$$\text{HCF}(12, 20) = 4, \quad \text{HCF}(12, 28) = 4 \Rightarrow n = 20 \text{ or } 28$$

(The "|" sign means "divides exactly": $4 \mid 20$ reads "4 divides 20".)

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Place value	each column is worth 10 times the column to its right; 0 holds an empty place
Adding a negative	$a + (-b) = a - b$ e.g. $7 + (-3) = 4$
Subtracting a negative	$a - (-b) = a + b$ e.g. $5 - (-2) = 7$
Different signs (+, −)	subtract the sizes; the answer takes the sign of the number with the bigger size: $-9 + 4 = -5$
× and ÷ signs	$(-a) \times (-b) = ab, \quad (-a) \times b = -ab$ same signs give $+$, different signs give $-$; the same for $\div$
Order of operations	brackets (innermost first), powers, × and ÷ left to right, + and − left to right
Fraction bar	$\dfrac{a+b}{c+d} = (a+b) \div (c+d)$ the bar acts as brackets
Powers of negatives	$(-3)^2 = 9, \qquad -3^2 = -9$
Commutative	$a + b = b + a, \quad a \times b = b \times a$ **not** for $-$ or $\div$
Distributive law	$a(b + c) = ab + ac$
Division	$a \times b = c \iff c \div b = a$ never divide by 0
Remainder	number = quotient × divisor + remainder (remainder < divisor)
Prime factorisation	every whole number above 1 is one product of primes, e.g. $60 = 2^2 \times 3 \times 5$
HCF	primes common to all the numbers, lowest powers
LCM	every prime that appears, highest powers
HCF and LCM	$\mathrm{HCF}(a, b) \times \mathrm{LCM}(a, b) = a \times b$
Find an unknown	undo the operation with its inverse, then check

Number detective: your turn

Try each one on paper or straight onto this page. Answers are underneath each group, written small and upside-down... well, small and grey. Cover them with your hand!

Bundles

1. What does the 6 in 3602 stand for?
2. 68 + 57
3. 81 - 36

1. 6 hundreds (600) 2. 125 (ones: 8+7 = 15, carry 1) 3. 45 (open a bundle: 11 - 6, 7 - 3)

The lift

4. -4 + 9
5. 3 - 8
6. -2 - (-5)
7. (-4) x (-5)

4. 5 5. -5 6. 3 (taking away a debt) 7. 20 (same signs: positive)

Getting dressed

8. 10 - 2 x 3
9. (10 - 2) x 3
10. $2 + 3^2$

8. 4 (multiply first) 9. 24 (brackets first) 10. 11 (power first: 9, then + 2)

Dots and lights

11. All factors of 18
12. Is 21 prime?
13. LCM of 6 and 8
14. HCF of 24 and 36

11. 1, 2, 3, 6, 9, 18 12. No: 21 = 3 x 7 13. 24 14. 12

Signs, order and primes

15. $-6 + (-9)$
16. $-12 \div 3 \times 2$
17. $(-5)^2$ and -5^2
18. $2 \times [10 - (1 + 2)^2]$
19. $(18 - 3) / (2 + 3)$
20. HCF and LCM of 18 and 24, using prime factors

15. -15 (both negative: add sizes) 16. -8 (left to right: -4×2) 17. 25 and -25 18. 2 (inner bracket 3, then $10 - 9 = 1$) 19. 3 ($15 \div 5$) 20. $18 = 2 \times 3^2$, $24 = 2^3 \times 3$: HCF $= 2 \times 3 = 6$, LCM $= 2^3 \times 3^2 = 72$

Find the unknown

21. $? - 7 = -3$
22. $-4 \times ? = 28$
23. $? \div (-6) = 5$
24. $15 + ? = 9$
25. The HCF of 12 and n is 4, and n is from 20 to 30. Find n.
26. The LCM of 6 and n is 30, and n is less than 10. Find n.

21. 4 ($? = -3 + 7$) 22. -7 ($? = 28 \div (-4)$) 23. -30 ($? = 5 \times (-6)$) 24. -6 ($? = 9 - 15$) 25. 20 or 28 (multiples of 4; 24 is out because HCF(12, 24) = 12) 26. 5 ($30 = 2 \times 3 \times 5$ and $6 = 2 \times 3$, so n must supply the 5; 10 is too big)